MACHINE LEARNING - REGRESSION

Online News Popularity

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1. Introduction

Online news platforms rely on user engagement and gather information regarding user behavioural patterns. One of the most widely used indicators of engagement is the number of times an article is shared on social media. Being able to predict an article's popularity before publication is valuable for media companies, editors and content strategists, yet it remains a challenging task due to the complex and often unpredictable nature of human behavior.

In this project, we formulate **a supervised regression problem** aimed at predicting the number of shares an online news article will receive. We use the **Online News Popularity dataset**, publicly available from the **UCI Machine Learning Repository** (link), which contains information on nearly 40,000 news articles published by *Mashable*. The dataset provides a rich set of features describing article content, sentiment, timing and structure.

The target variable, (as suggested) `shares`, represents the total number of times an article was shared across multiple social media platforms. The feature set includes: statistics related to word usage, sentiment polarity, presence of multimedia content, keyword metrics and publication day.

The main objectives of this project are:

- To explore and preprocess the dataset in a reproducible and well-documented manner,
- To train and tune multiple regression models with different levels of complexity,
- To compare model performance using appropriate regression metrics,
- To analyze prediction errors and understand model limitations.

To address these objectives, we experiment with **five regression algorithms**.

2. Data Preprocessing

We start our project with loading the libraries.

In [99]:
```python
# Libraries import
import pandas as pd
from IPython.display import display
import matplotlib.pyplot as plt
import numpy as np
from sklearn.model_selection import train_test_split
import math
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.ensemble import RandomForestRegressor
from sklearn.metrics import mean_squared_error, r2_score, make_scorer
from sklearn.model_selection import RandomizedSearchCV
import matplotlib.pyplot as plt
from sklearn.linear_model import ElasticNet
from sklearn.ensemble import GradientBoostingRegressor
from sklearn.svm import SVR
from xgboost import XGBRegressor
```

Later, we cover the file path, in the way the code would be reproducible for other users.

In [100…
```python
csv_path = 'OnlineNewsPopularity.csv'

df = pd.read_csv(csv_path)
print(f'Loaded local file: {csv_path}')
display(df.head())
```

```
Loaded local file: OnlineNewsPopularity.csv
```

	url	timedelta	n_tokens_title	n_tokens_content	n_unique_tokens	n_non_stop_words	n_non_stop_unique_tokens	num_hrefs	num_self_hrefs	num_imgs	…	min_
0	http://mashable.com/2013/01/07/amazon-instant-...	731.0	12.0	219.0	0.663594	1.0	0.815385	4.0	2.0	1.0	…	
1	http://mashable.com/2013/01/07/ap-samsung-spon...	731.0	9.0	255.0	0.604743	1.0	0.791946	3.0	1.0	1.0	…	
2	http://mashable.com/2013/01/07/apple-40-billio...	731.0	9.0	211.0	0.575130	1.0	0.663866	3.0	1.0	1.0	…	
3	http://mashable.com/2013/01/07/astronaut-notre...	731.0	9.0	531.0	0.503788	1.0	0.665635	9.0	0.0	1.0	…	
4	http://mashable.com/2013/01/07/att-u-verse-apps/	731.0	13.0	1072.0	0.415646	1.0	0.540890	19.0	19.0	20.0	…	

5 rows × 61 columns

In [101…
```python
# Show column names
cols = df.columns.tolist()
print("Number of columns:", len(cols))
for c in cols:
    print(repr(c))
```

```
Number of columns: 61
'url'
' timedelta'
' n_tokens_title'
' n_tokens_content'
' n_unique_tokens'
' n_non_stop_words'
' n_non_stop_unique_tokens'
' num_hrefs'
' num_self_hrefs'
' num_imgs'
' num_videos'
' average_token_length'
' num_keywords'
' data_channel_is_lifestyle'
' data_channel_is_entertainment'
' data_channel_is_bus'
' data_channel_is_socmed'
' data_channel_is_tech'
' data_channel_is_world'
' kw_min_min'
' kw_max_min'
' kw_avg_min'
' kw_min_max'
' kw_max_max'
' kw_avg_max'
' kw_min_avg'
' kw_max_avg'
' kw_avg_avg'
' self_reference_min_shares'
' self_reference_max_shares'
' self_reference_avg_sharess'
' weekday_is_monday'
' weekday_is_tuesday'
' weekday_is_wednesday'
' weekday_is_thursday'
' weekday_is_friday'
' weekday_is_saturday'
' weekday_is_sunday'
' is_weekend'
' LDA_00'
' LDA_01'
' LDA_02'
' LDA_03'
' LDA_04'
' global_subjectivity'
' global_sentiment_polarity'
' global_rate_positive_words'
' global_rate_negative_words'
' rate_positive_words'
' rate_negative_words'
' avg_positive_polarity'
' min_positive_polarity'
' max_positive_polarity'
' avg_negative_polarity'
' min_negative_polarity'
' max_negative_polarity'
' title_subjectivity'
' title_sentiment_polarity'
' abs_title_subjectivity'
' abs_title_sentiment_polarity'
' shares'
```

The dataset contains **61 variables**, including one target variable (`shares`) and a rich set of features describing article content, metadata, publication timing, and sentiment. For clarity, the features are grouped into meaningful categories below.

1. Identifiers and Non-Predictive Features

- **`url`** : URL of the article (identifier, non-predictive)
- **`timedelta`** : Number of days between article publication and dataset acquisition (non-predictive)

2. Content Structure and Length Features

- **`n_tokens_title`** : Number of words in the article title
- **`n_tokens_content`** : Number of words in the article content
- **`average_token_length`** : Average word length in the content
- **`num_keywords`** : Number of keywords in the article metadata

3. Lexical Diversity and Text Composition

- **`n_unique_tokens`** : Proportion of unique words in the content
- **`n_non_stop_words`** : Proportion of non-stop words
- **`n_non_stop_unique_tokens`** : Proportion of unique non-stop words

4. Hyperlinks and Multimedia Content

- **`num_hrefs`** : Number of external links in the article
- **`num_self_hrefs`** : Number of links to other Mashable articles
- **`num_imgs`** : Number of images
- **`num_videos`** : Number of embedded videos

5. Data Channel Indicators

Binary variables indicating the article's category:

- **`data_channel_is_lifestyle`**
- **`data_channel_is_entertainment`**
- **`data_channel_is_bus`**
- **`data_channel_is_socmed`**
- **`data_channel_is_tech`**
- **`data_channel_is_world`**

6. Keyword Popularity Statistics

Statistics describing the popularity of keywords associated with the article:

- **Worst-performing keywords**:
  - `kw_min_min` , `kw_max_min` , `kw_avg_min`
- **Best-performing keywords**:
  - `kw_min_max` , `kw_max_max` , `kw_avg_max`
- **Average keyword performance**:
  - `kw_min_avg` , `kw_max_avg` , `kw_avg_avg`

7. Self-Reference Popularity Features

- **`self_reference_min_shares`** : Minimum shares of referenced Mashable articles
- **`self_reference_max_shares`** : Maximum shares of referenced Mashable articles
- **`self_reference_avg_sharess`** : Average shares of referenced Mashable articles

8. Publication Timing Features

- **Weekday indicators**:
  - `weekday_is_monday` , `weekday_is_tuesday` , `weekday_is_wednesday` , `weekday_is_thursday` , `weekday_is_friday` , `weekday_is_saturday` , `weekday_is_sunday`

9. Topic Distribution (LDA Features)

- **`LDA_00` – `LDA_04`** : Degree of association with five latent topics derived using Latent Dirichlet Allocation (LDA)

10. Sentiment and Subjectivity Features

- **`global_subjectivity`** : Overall text subjectivity
- **`global_sentiment_polarity`** : Overall sentiment polarity
- **`global_rate_positive_words`** : Proportion of positive words
- **`global_rate_negative_words`** : Proportion of negative words
- **`rate_positive_words`** : Proportion of positive words among non-neutral tokens
- **`rate_negative_words`** : Proportion of negative words among non-neutral tokens
- `avg_positive_polarity` , `min_positive_polarity` , `max_positive_polarity`
- `avg_negative_polarity` , `min_negative_polarity` , `max_negative_polarity`

11. Sentiment and Subjectivity Features (Title)

- **`title_subjectivity`** : Subjectivity of the article title
- **`title_sentiment_polarity`** : Sentiment polarity of the title
- **`abs_title_subjectivity`** : Absolute subjectivity level of the title
- **`abs_title_sentiment_polarity`** : Absolute sentiment polarity level of the title

12. Target Variable

- **`shares`** : Number of times the article was shared on social media platforms

In [102…
```python
# Remove leading spaces from column names
df.columns = df.columns.str.strip()
```

Now, basic checks and checking the nulll values.

In [103…
```python
# Basic checks
print("Shape:", df.shape)
print("\nColumns:", df.columns.tolist())
print("\nData types:\n", df.dtypes)

# Missing values
missing = df.isna().sum().sort_values(ascending=False)
print("\nMissing values (top):\n", missing.head(10))

# Duplicates
print("\nDuplicate rows:", df.duplicated().sum())
```

```
Shape: (39644, 61)

Columns: ['url', 'timedelta', 'n_tokens_title', 'n_tokens_content', 'n_unique_tokens', 'n_non_stop_words', 'n_non_stop_unique_tokens', 'num_hrefs', 'num_self_hrefs', 'num_imgs',
'num_videos', 'average_token_length', 'num_keywords', 'data_channel_is_lifestyle', 'data_channel_is_entertainment', 'data_channel_is_bus', 'data_channel_is_socmed', 'data_channel
_is_tech', 'data_channel_is_world', 'kw_min_min', 'kw_max_min', 'kw_avg_min', 'kw_min_max', 'kw_max_max', 'kw_avg_max', 'kw_min_avg', 'kw_max_avg', 'kw_avg_avg', 'self_reference_
min_shares', 'self_reference_max_shares', 'self_reference_avg_sharess', 'weekday_is_monday', 'weekday_is_tuesday', 'weekday_is_wednesday', 'weekday_is_thursday', 'weekday_is_frid
ay', 'weekday_is_saturday', 'weekday_is_sunday', 'is_weekend', 'LDA_00', 'LDA_01', 'LDA_02', 'LDA_03', 'LDA_04', 'global_subjectivity', 'global_sentiment_polarity', 'global_rate_
positive_words', 'global_rate_negative_words', 'rate_positive_words', 'rate_negative_words', 'avg_positive_polarity', 'min_positive_polarity', 'max_positive_polarity', 'avg_negat
ive_polarity', 'min_negative_polarity', 'max_negative_polarity', 'title_subjectivity', 'title_sentiment_polarity', 'abs_title_subjectivity', 'abs_title_sentiment_polarity', 'shar
es']

Data types:
 url                             object
timedelta                      float64
n_tokens_title                 float64
n_tokens_content               float64
n_unique_tokens                float64
                                ...
title_subjectivity             float64
title_sentiment_polarity       float64
abs_title_subjectivity         float64
abs_title_sentiment_polarity   float64
shares                           int64
Length: 61, dtype: object

Missing values (top):
 url                      0
weekday_is_monday       0
weekday_is_wednesday    0
weekday_is_thursday     0
weekday_is_friday       0
weekday_is_saturday     0
weekday_is_sunday       0
is_weekend              0
LDA_00                  0
LDA_01                  0
dtype: int64

Duplicate rows: 0
```

We confirmed, what authors of the dataset say, there are no missing values.

3. Explanatory Data Analysis and Feature Engineering

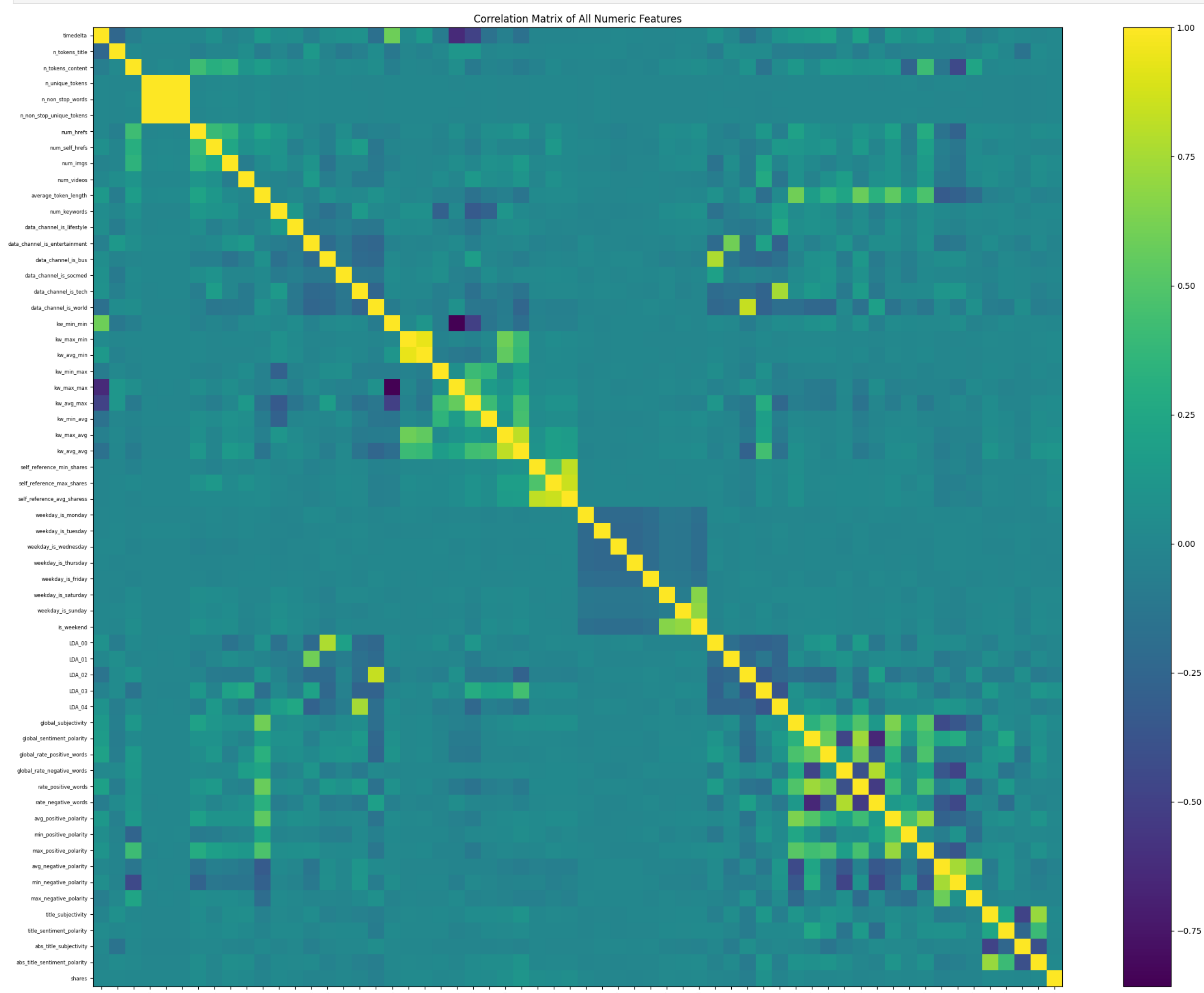
Correlation Heatmap

Let's check the correlations between the target variable and the features.

In [104…
```python
corr_matrix = df.corr(numeric_only=True)

# Correlation with target
target_corr = corr_matrix["shares"].sort_values(ascending=False)

display(target_corr.head(10))
display(target_corr.tail(10))
```

```
shares                        1.000000
kw_avg_avg                    0.110413
LDA_03                        0.083771
kw_max_avg                    0.064306
self_reference_avg_sharess    0.057789
self_reference_min_shares     0.055958
self_reference_max_shares     0.047115
num_hrefs                     0.045404
kw_avg_max                    0.044686
kw_min_avg                    0.039551
Name: shares, dtype: float64
rate_positive_words             -0.013241
data_channel_is_tech            -0.013253
LDA_04                          -0.016622
data_channel_is_entertainment   -0.017006
min_negative_polarity           -0.019297
max_negative_polarity           -0.019300
average_token_length            -0.022007
avg_negative_polarity           -0.032029
data_channel_is_world           -0.049497
LDA_02                          -0.059163
Name: shares, dtype: float64
```

In [105…
```python
plt.figure(figsize=(22, 18))
plt.imshow(corr_matrix, aspect="auto")
plt.colorbar(label="Correlation coefficient")
plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns, rotation=90, fontsize=6)
plt.yticks(range(len(corr_matrix.columns)), corr_matrix.columns, fontsize=6)
plt.title("Correlation Matrix of All Numeric Features")
plt.tight_layout()
plt.show()
```


Correlation Matrix of All Numeric Features

In [106…
```python
df["kw_worst_mean"] = df[["kw_min_min", "kw_max_min", "kw_avg_min"]].mean(axis=1)
df["kw_best_mean"]  = df[["kw_min_max", "kw_max_max", "kw_avg_max"]].mean(axis=1)
df["kw_avg_mean"]   = df[["kw_min_avg", "kw_max_avg", "kw_avg_avg"]].mean(axis=1)

df = df.drop(columns=[
    "kw_min_min", "kw_max_min", "kw_avg_min",
    "kw_min_max", "kw_max_max", "kw_avg_max",
    "kw_min_avg", "kw_max_avg", "kw_avg_avg"
])
```

In [107…
```python
df["self_reference_mean_shares"] = df[
    ["self_reference_min_shares",
     "self_reference_max_shares",
     "self_reference_avg_sharess"]
].mean(axis=1)

df = df.drop(columns=[
    "self_reference_min_shares",
    "self_reference_max_shares",
    "self_reference_avg_sharess"
])
```

In [108…
```python
df["content_sentiment_strength"] = (
    df["global_rate_positive_words"] +
    df["global_rate_negative_words"]
)

df = df.drop(columns=[
    "global_rate_positive_words",
    "global_rate_negative_words"
])

df["content_polarity_range"] = (
    df["max_positive_polarity"] - df["min_negative_polarity"]
)

df = df.drop(columns=[
    "max_positive_polarity",
    "min_negative_polarity"
])
```

In [109…
```python
n_rows, n_cols = df.shape
print(f"Dataset shape after aggregation: {n_rows} rows, {n_cols} columns")
```

```
Dataset shape after aggregation: 39644 rows, 51 columns
```

In [110…
```python
# Identify weekday dummy columns
weekday_cols = [c for c in df.columns if c.startswith("weekday_is_")]

# Create categorical weekday variable
df["weekday"] = (
    df[weekday_cols]
    .idxmax(axis=1)
    .str.replace("weekday_is_", "")
)

# Drop redundant columns
df = df.drop(columns=weekday_cols + ["is_weekend"])
```

In [111…
```python
print(df["weekday"].value_counts())
print("Number of columns after weekday removal:", df.shape[1])
```

```
weekday
wednesday    7435
tuesday      7390
thursday     7267
monday       6661
friday       5701
sunday       2737
saturday     2453
Name: count, dtype: int64
Number of columns after weekday removal: 44
```

In [112…
```python
channel_cols = [
    "data_channel_is_lifestyle",
    "data_channel_is_entertainment",
    "data_channel_is_bus",
    "data_channel_is_socmed",
    "data_channel_is_tech",
    "data_channel_is_world"
]
```

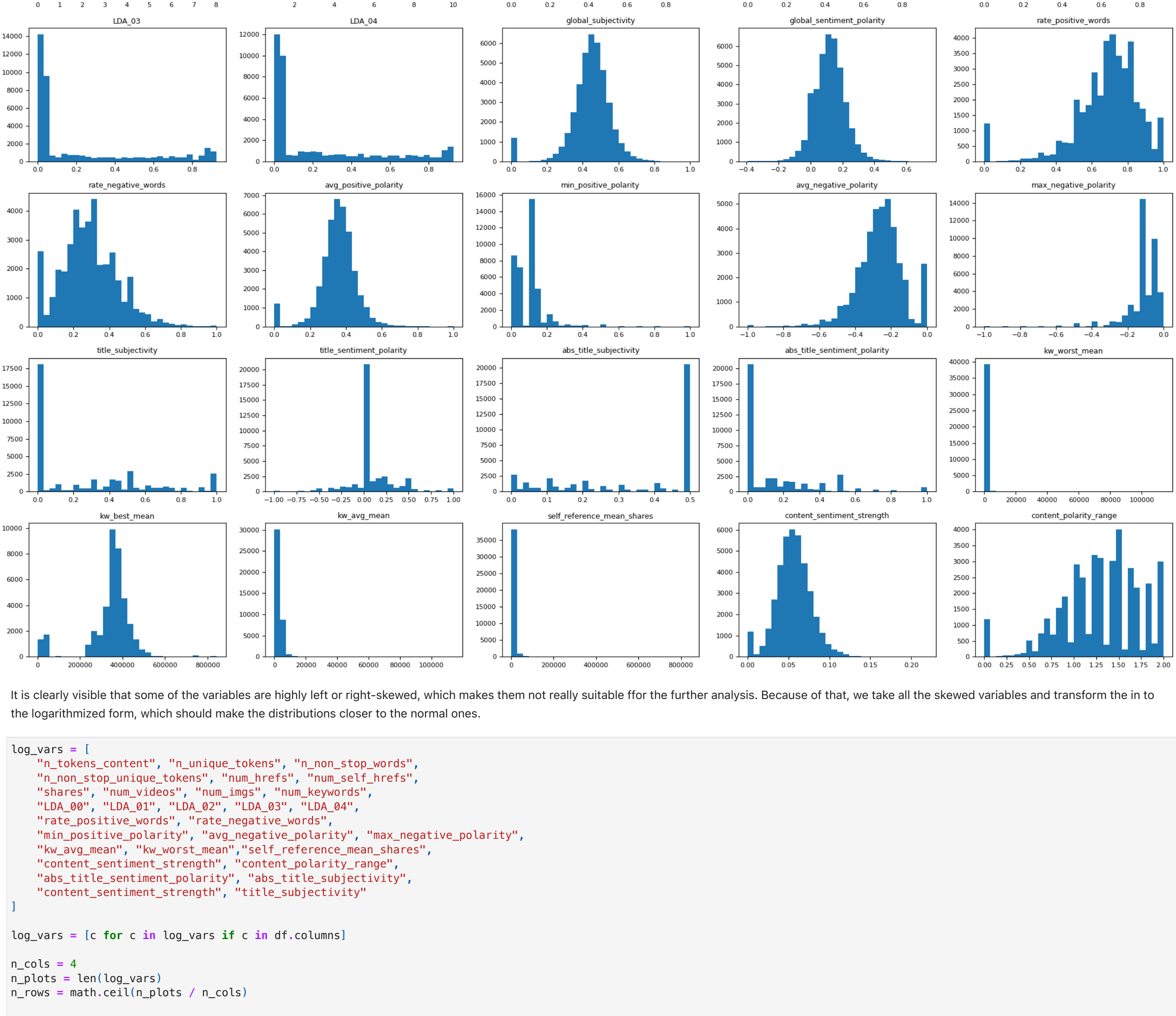
In [113…
```python
df["data_channel"] = (
    df[channel_cols]
    .idxmax(axis=1)
    .str.replace("data_channel_is_", "")
)
```

In [114…
```python
# Drop redundant columns
df = df.drop(columns=channel_cols)
```

In [115…
```python
df["data_channel"].value_counts()
```

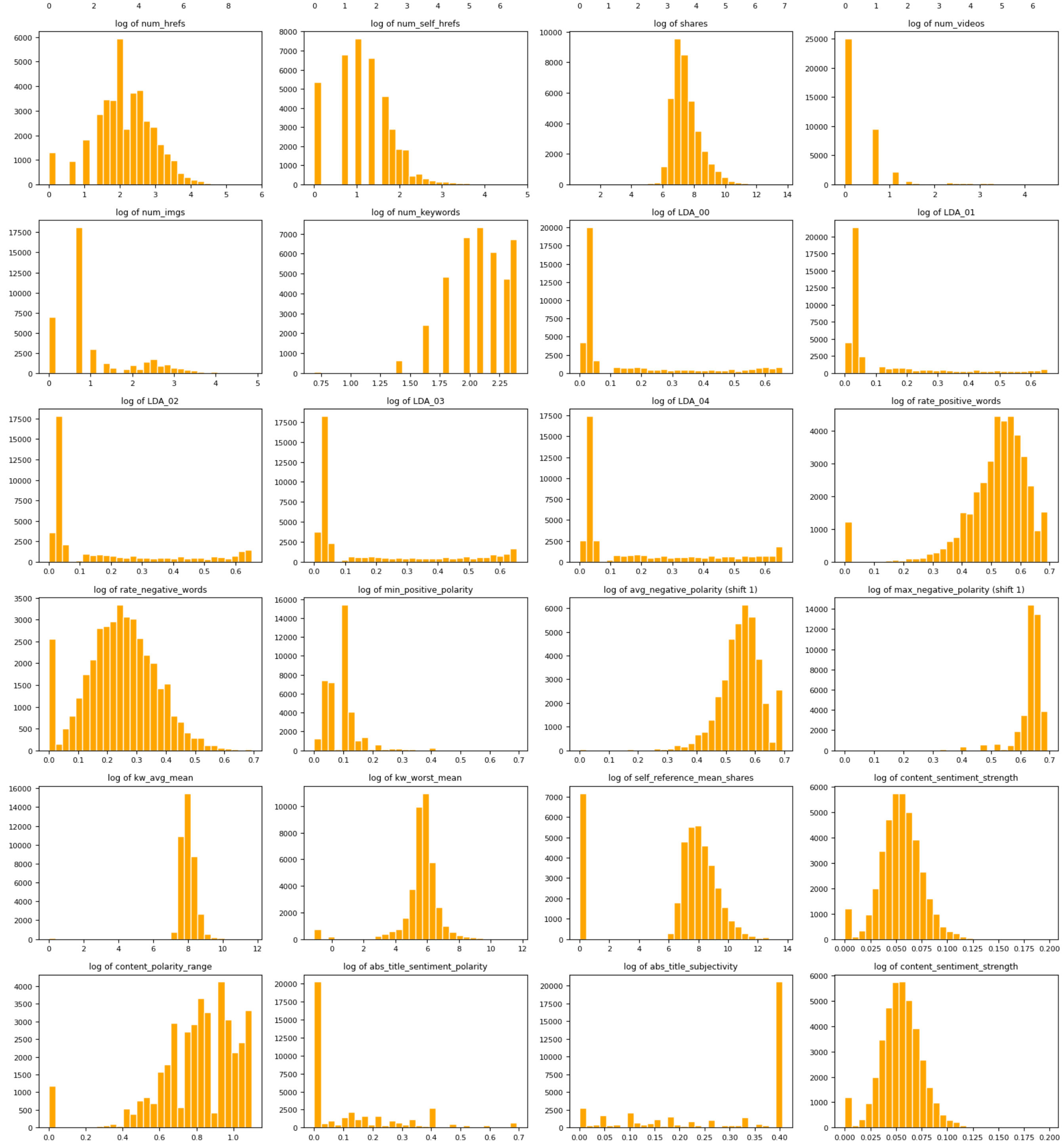
Out[115…
```
data_channel
world            8427
lifestyle        8233
tech             7346
entertainment    7057
bus              6258
socmed           2323
Name: count, dtype: int64
```

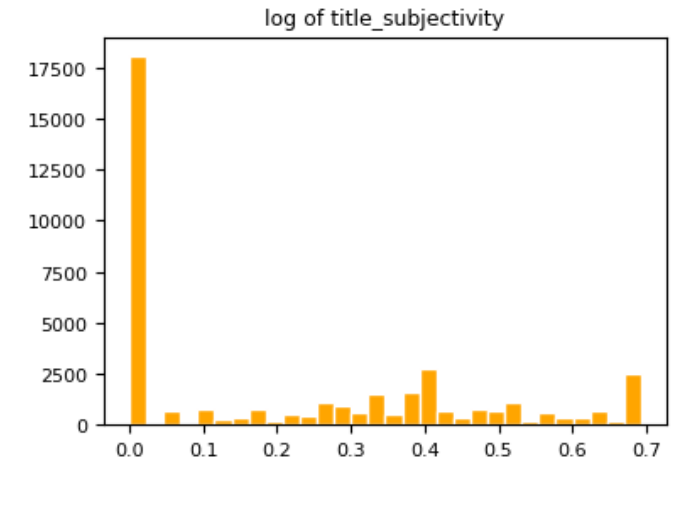
In [116…
```python
# Select numeric columns only, exclude target
numeric_cols = df.select_dtypes(include=["number"]).columns.tolist()
numeric_cols = [c for c in numeric_cols if c != "shares"]  # predictors only

# Grid layout
n_cols = 5
n_plots = len(numeric_cols)
n_rows = math.ceil(n_plots / n_cols)

# Create the figure
fig, axes = plt.subplots(n_rows, n_cols, figsize=(4*n_cols, 2.8*n_rows))
axes = axes.flatten()

# Plot each histogram
for i, col in enumerate(numeric_cols):
    axes[i].hist(df[col].dropna(), bins=30)
    axes[i].set_title(col, fontsize=9)
    axes[i].tick_params(axis="both", labelsize=8)

# Hide unused subplots (if any)
for j in range(n_plots, len(axes)):
    axes[j].axis("off")

plt.tight_layout()
plt.show()
```


timedelta
n_tokens_title
n_tokens_content
n_unique_tokens
n_non_stop_words
n_non_stop_unique_tokens
num_hrefs
num_self_hrefs
num_imgs
num_videos
average_token_length
num_keywords
LDA_00
LDA_01
LDA_02
LDA_03
LDA_04
global_subjectivity
global_sentiment_polarity
rate_positive_words
rate_negative_words
avg_positive_polarity
min_positive_polarity
avg_negative_polarity
max_negative_polarity
title_subjectivity
title_sentiment_polarity
abs_title_subjectivity
abs_title_sentiment_polarity
kw_worst_mean
kw_best_mean
kw_avg_mean
self_reference_mean_shares
content_sentiment_strength
content_polarity_range

It is clearly visible that some of the variables are highly left or right-skewed, which makes them not really suitable ffor the further analysis. Because of that, we take all the skewed variables and transform the in to the logarithmized form, which should make the distributions closer to the normal ones.

In [117…
```python
log_vars = [
    "n_tokens_content", "n_unique_tokens", "n_non_stop_words",
    "n_non_stop_unique_tokens", "num_hrefs", "num_self_hrefs",
    "shares", "num_videos", "num_imgs", "num_keywords",
    "LDA_00", "LDA_01", "LDA_02", "LDA_03", "LDA_04",
    "rate_positive_words", "rate_negative_words",
    "min_positive_polarity", "avg_negative_polarity", "max_negative_polarity",
    "kw_avg_mean", "kw_worst_mean","self_reference_mean_shares",
    "content_sentiment_strength", "content_polarity_range",
    "abs_title_sentiment_polarity", "abs_title_subjectivity",
    "content_sentiment_strength", "title_subjectivity"
]

log_vars = [c for c in log_vars if c in df.columns]

n_cols = 4
n_plots = len(log_vars)
n_rows = math.ceil(n_plots / n_cols)

fig, axes = plt.subplots(n_rows, n_cols, figsize=(4*n_cols, 3*n_rows))
axes = np.array(axes).flatten()

for i, col in enumerate(log_vars):
    raw = df[col].dropna()

    # Shift if needed so all values are > −1 for log1p
    min_val = raw.min()
    shift = (−min_val + 1e−6) if min_val <= −1 else 0.0

    data = np.log1p(raw + shift)

    axes[i].hist(data, bins=30, color="orange", edgecolor="white")
    if shift > 0:
        axes[i].set_title(f"log of {col} (shift {shift:.3g})", fontsize=9)
    else:
        axes[i].set_title(f"log of {col}", fontsize=9)
    axes[i].tick_params(axis="both", labelsize=8)

for j in range(n_plots, len(axes)):
    axes[j].axis("off")

plt.suptitle("Histograms of Log−Transformed Variables", fontsize=14)
plt.tight_layout(rect=[0, 0, 1, 0.96])
plt.show()
```


Histograms of Log-Transformed Variables
log of n_tokens_content
log of n_unique_tokens
log of n_non_stop_words
log of n_non_stop_unique_tokens
log of num_hrefs
log of num_self_hrefs
log of shares
log of num_videos
log of num_imgs
log of num_keywords
log of LDA_00
log of LDA_01
log of LDA_02
log of LDA_03
log of LDA_04
log of rate_positive_words
log of rate_negative_words
log of min_positive_polarity
log of avg_negative_polarity (shift 1)
log of max_negative_polarity (shift 1)
log of kw_avg_mean
log of kw_worst_mean
log of self_reference_mean_shares
log of content_sentiment_strength
log of content_polarity_range
log of abs_title_sentiment_polarity
log of abs_title_subjectivity
log of content_sentiment_strength



After this operation it's visible that some of the features are still not performing well in term of normal distribution. It is time to get rid of them.

In [118…

```python
# Drop specified (possibly already-removed) columns safely and report outcome
cols_to_drop = [
    "url",
    "timedelta",
    "n_unique_tokens",
    "n_non_stop_words",
    "n_non_stop_unique_tokens",
    "num_videos",
    "num_imgs",
    "num_self_hrefs",
    "LDA_00",
    "LDA_01",
    "LDA_02",
    "LDA_03",
    "LDA_04",
    "min_positive_polarity",
    "max_negative_polarity",
    "rate_negative_words",
    "content_polarity_range",
    "abs_title_sentiment_polarity",
    "abs_title_subjectivity",
    "title_subjectivity",
]



existing = [c for c in cols_to_drop if c in df.columns]
missing = [c for c in cols_to_drop if c not in df.columns]

if existing:
    df.drop(columns=existing, inplace=True, errors="ignore")

print(f"Dropped columns ({len(existing)}): {existing}")
if missing:
    print(f"Ignored (not present): {missing}")
print("New dataframe shape:", df.shape)
```

```
Dropped columns (20): ['url', 'timedelta', 'n_unique_tokens', 'n_non_stop_words', 'n_non_stop_unique_tokens', 'num_videos', 'num_imgs', 'num_self_hrefs', 'LDA_00', 'LDA_01', 'LDA
_02', 'LDA_03', 'LDA_04', 'min_positive_polarity', 'max_negative_polarity', 'rate_negative_words', 'content_polarity_range', 'abs_title_sentiment_polarity', 'abs_title_subjectivi
ty', 'title_subjectivity']
New dataframe shape: (39644, 19)
```

At this point, our dataset contains 19 columns.

Knowing which columns are performing well after logarithmization, we change them also in the dataframe.

In [119…

```python
# Create log-transformed columns for specified features
cols_to_log = [
    "n_tokens_content",
    "num_hrefs",
    "num_keywords",
    "rate_positive_words",
    "avg_negative_polarity",
    "kw_avg_mean",
    "kw_worst_mean",
    "self_reference_mean_shares",
    "content_sentiment_strength"
]

created = []
skipped = []

for col in cols_to_log:
    if col not in df.columns:
        skipped.append(col)
        continue

    # shift if necessary so values > -1 for log1p
    min_val = df[col].min()
    shift = (-min_val + 1e-6) if min_val <= -1 else 0.0

    new_col = f"log_{col}"
    df[new_col] = np.log1p(df[col] + shift)
    created.append((new_col, shift))

# Report
if created:
    print("Created log columns:")
    for new_col, shift in created:
        if shift:
            print(f" - {new_col} (shifted by {shift:.6g} before log1p)")
        else:
            print(f" - {new_col}")
if skipped:
    print("Skipped (not found):", skipped)
```

```
Created log columns:
 - log_n_tokens_content
 - log_num_hrefs
 - log_num_keywords
 - log_rate_positive_words
 - log_avg_negative_polarity (shifted by 1 before log1p)
 - log_kw_avg_mean
 - log_kw_worst_mean
 - log_self_reference_mean_shares
 - log_content_sentiment_strength
```

Here, we drop the columns, which logarithms we created.

In [85]:

```python
# Drop specified columns safely
cols_to_drop = [
    "n_tokens_content",
    "num_hrefs",
    "num_keywords",
    "rate_positive_words",
    "avg_negative_polarity",
    "kw_avg_mean",
    "kw_worst_mean",
    "self_reference_mean_shares",
    "content_sentiment_strength"
]

existing = [c for c in cols_to_drop if c in df.columns]
missing = [c for c in cols_to_drop if c not in df.columns]

if existing:
    df.drop(columns=existing, inplace=True, errors="ignore")

print(f"Dropped ({len(existing)}): {existing}")
if missing:
    print(f"Not present (skipped): {missing}")
print("New dataframe shape:", df.shape)

# Update common helper lists in the notebook to avoid references to removed columns
for var in ["numeric_cols", "numeric_features", "other_numeric", "log_features", "log_feats", "predictors", "cols"]:
    if var in globals() and isinstance(globals()[var], list):
        globals()[var] = [x for x in globals()[var] if x not in cols_to_drop]
```

```
Dropped (9): ['n_tokens_content', 'num_hrefs', 'num_keywords', 'rate_positive_words', 'avg_negative_polarity', 'kw_avg_mean', 'kw_worst_mean', 'self_reference_mean_shares', 'cont
ent_sentiment_strength']
New dataframe shape: (39644, 19)
```

Let's do a quick check of the dataframe.

In [86]:

```python
df
```

Out[86]:

	n_tokens_title	average_token_length	global_subjectivity	global_sentiment_polarity	avg_positive_polarity	title_sentiment_polarity	shares	kw_best_mean	weekday	data_channel
0	12.0	4.680365	0.521617	0.092562	0.378636	-0.187500	593	0.000000	monday	entertainment
1	9.0	4.913725	0.341246	0.148948	0.286915	0.000000	711	0.000000	monday	bus
2	9.0	4.393365	0.702222	0.323333	0.495833	0.000000	1500	0.000000	monday	bus
3	9.0	4.404896	0.429850	0.100705	0.385965	0.000000	1200	0.000000	monday	entertainment
4	13.0	4.682836	0.513502	0.281003	0.411127	0.136364	505	0.000000	monday	tech
...	...	...	...	...	...	...	...	...	...	...
39639	11.0	4.523121	0.482679	0.141964	0.333791	0.000000	1800	415054.166667	wednesday	tech
39640	12.0	4.405488	0.564374	0.194249	0.374825	1.000000	1900	347595.238095	wednesday	socmed
39641	10.0	5.076923	0.510296	0.024609	0.307273	0.136364	1900	381783.333333	wednesday	lifestyle
39642	6.0	4.975073	0.358578	-0.008066	0.236851	0.000000	1100	365966.666667	wednesday	world
39643	10.0	4.471338	0.517893	0.104892	0.247338	0.250000	1300	471700.000000	wednesday	entertainment

39644 rows × 19 columns

Target Variable Analysis

As `shares` is our target varibale, let's now focus on it.

In [87]:

```python
# Summary of target
display(df["shares"].describe())
```

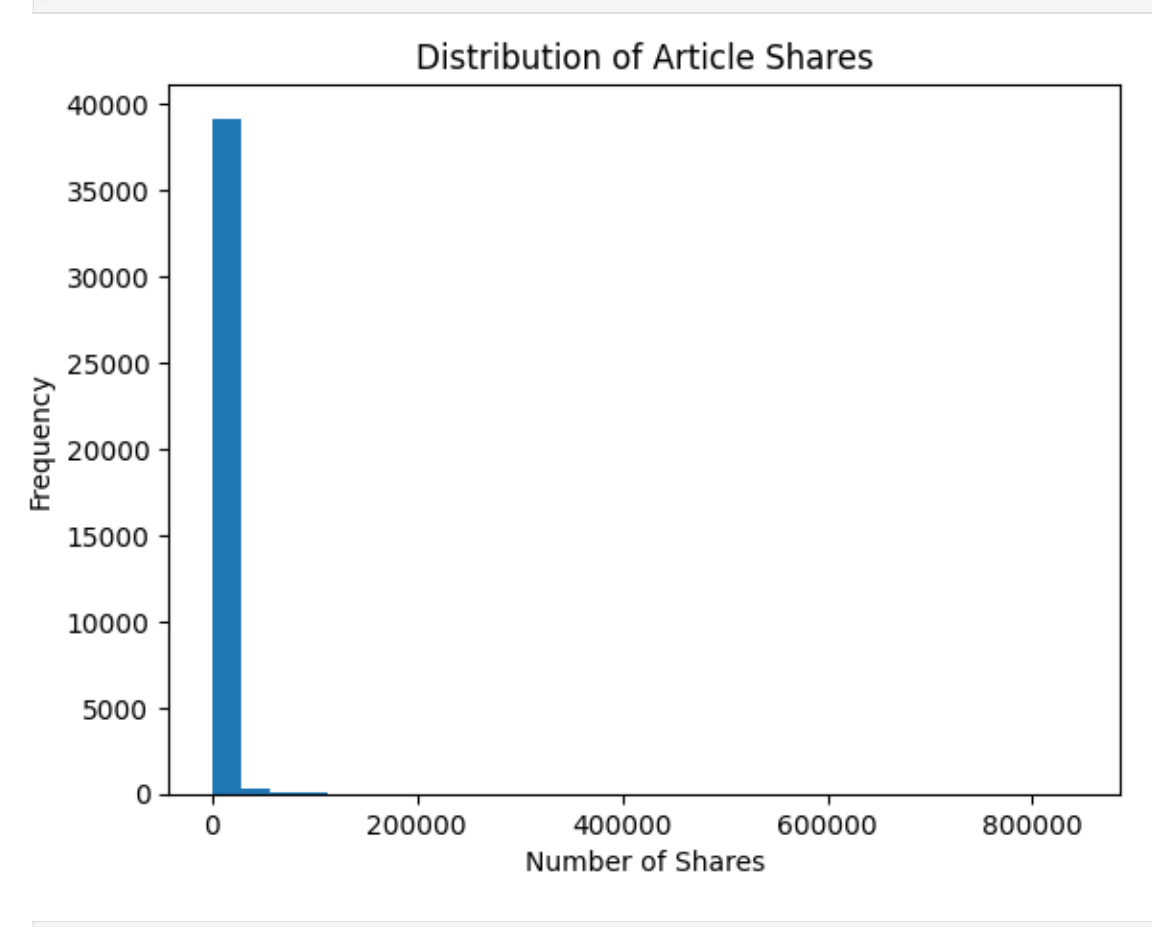
```
count     39644.000000
mean       3395.380184
std       11626.950749
min           1.000000
25%         946.000000
50%        1400.000000
75%        2800.000000
max      843300.000000
Name: shares, dtype: float64
```

- The target variable, `shares`, displays a **highly right-skewed distribution**.
- The **median** number of shares is **1400**, while the **mean** is substantially higher at **3395**, indicating the presence of extreme values.
- The **standard deviation (11,627)** is more than three times the mean, reflecting **very high variability** in article popularity.
- The **maximum observed value** reaches **843300 shares**, showing that a small number of articles achieve **viral popularity**, but they are perceived as outliers.
- In contrast, the majority of articles receive **relatively modest engagement**, with 75% of articles having fewer than **2800 shares**.

These characteristics motivate the use of a **logarithmic transformation of the target variable** in the modeling stage to reduce skewness and mitigate the influence of outliers.

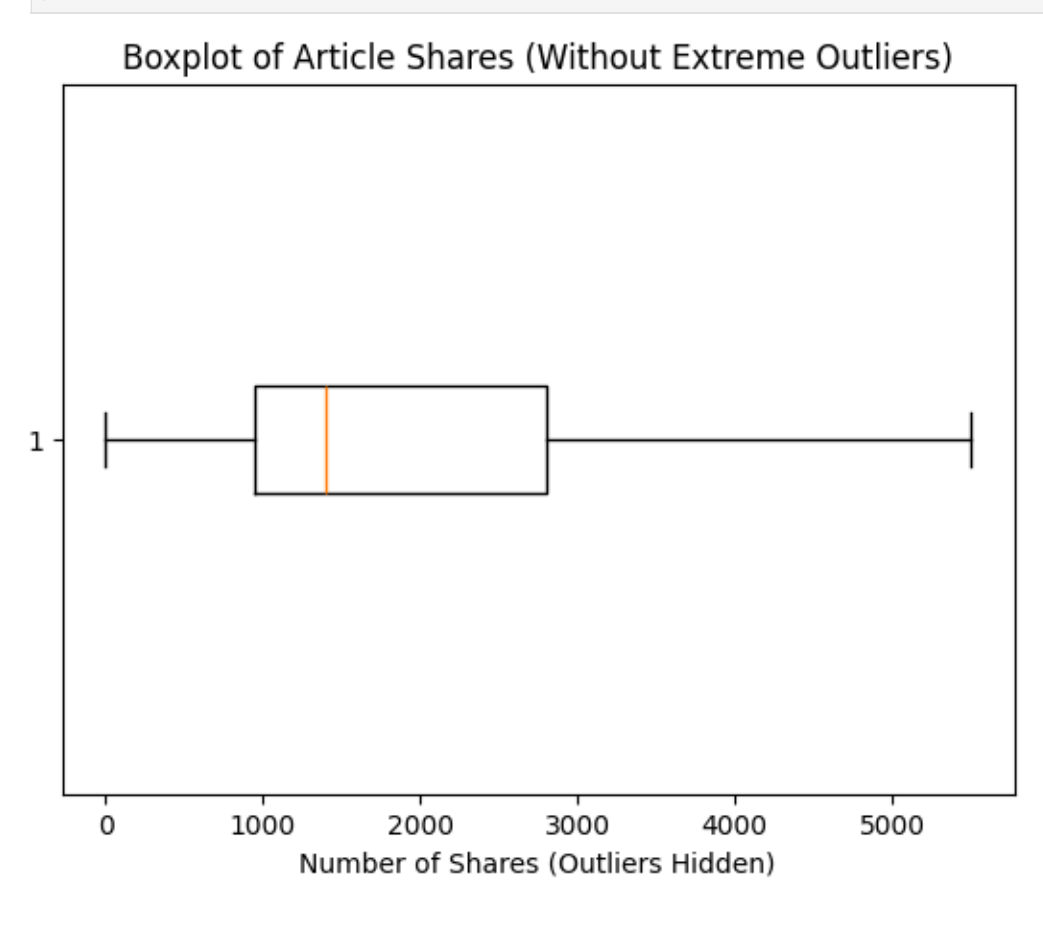
In [88]:

```python
plt.figure()
plt.hist(df["shares"], bins=30)
plt.xlabel("Number of Shares")
plt.ylabel("Frequency")
plt.title("Distribution of Article Shares")
plt.show()
```



In [89]:

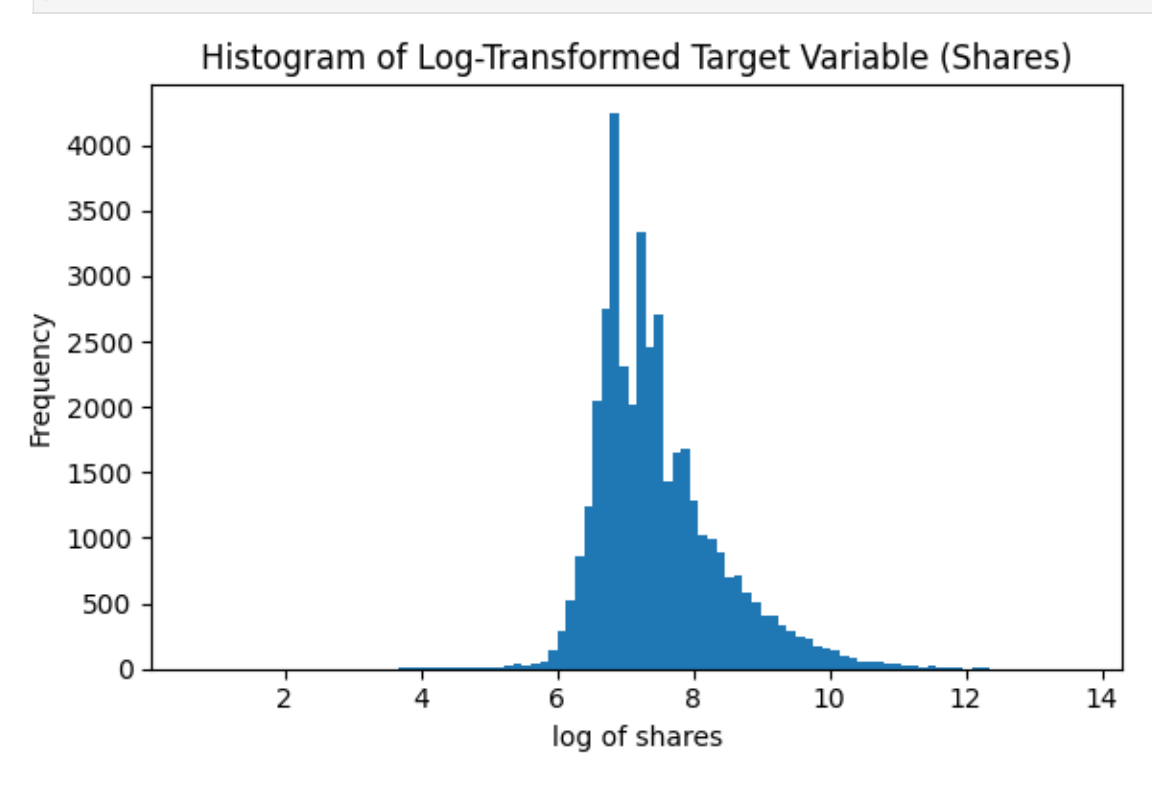
```python
plt.figure()
plt.boxplot(df["shares"], vert=False, showfliers=False)
plt.xlabel("Number of Shares (Outliers Hidden)")
plt.ylabel("Frequency")
plt.title("Boxplot of Article Shares (Without Extreme Outliers)")
plt.show()
```



The boxplot (with extreme outliers removed) shows a strongly right-skewed distribution of shares.

In [90]:

```python
# Log-transform target for visualization
log_shares = np.log1p(df["shares"])

plt.figure(figsize=(6, 4))
plt.hist(log_shares, bins=100)
plt.xlabel("log of shares")
plt.ylabel("Frequency")
plt.title("Histogram of Log-Transformed Target Variable (Shares)")
plt.tight_layout()
plt.show()
```



After the logarithmization the target reflects definitely more the normal distribution than before.

Next, we create the logarithmized `shares` in the dataset, and subsequently we remove the non-logarithmized `shares` from the dataframe.

In [91]:

```python
#create log-transformed target in the dataframe
df["log_shares"] = np.log1p(df["shares"])
```

In [92]:

```python
# Drop the raw target column 'shares' from the dataframe

if "shares" in df.columns:
    df.drop(columns=["shares"], inplace=True)
    print("Dropped 'shares'. New shape:", df.shape)
else:
    print("'shares' column not present; no action taken.")

# Quick check of remaining columns
print("Columns now:", df.columns.tolist())
```

```
Dropped 'shares'. New shape: (39644, 19)
Columns now: ['n_tokens_title', 'average_token_length', 'global_subjectivity', 'global_sentiment_polarity', 'avg_positive_polarity', 'title_sentiment_polarity', 'kw_best_mean', '
weekday', 'data_channel', 'log_n_tokens_content', 'log_num_hrefs', 'log_num_keywords', 'log_rate_positive_words', 'log_avg_negative_polarity', 'log_kw_avg_mean', 'log_kw_worst_me
an', 'log_self_reference_mean_shares', 'log_content_sentiment_strength', 'log_shares']
```

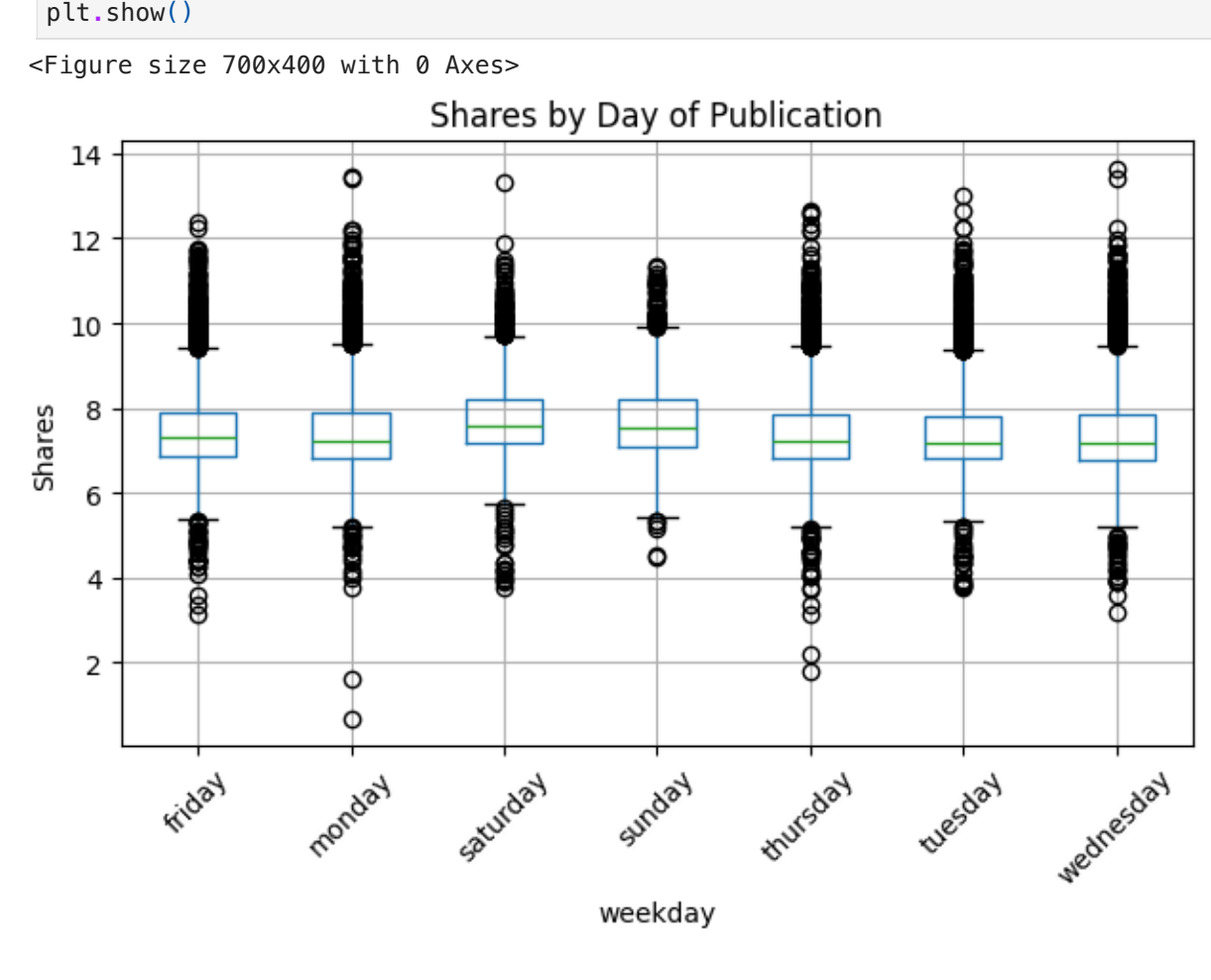
Predictor vs Target Variable

We can also check how the target variable is related to some of the features, for example how shares are changing withing a week.

In [93]:

```python
plt.figure(figsize=(7,4))
df.boxplot(column="log_shares", by="weekday")
plt.suptitle("")
plt.title("Shares by Day of Publication")
plt.ylabel("Shares")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```

```
<Figure size 700x400 with 0 Axes>
```



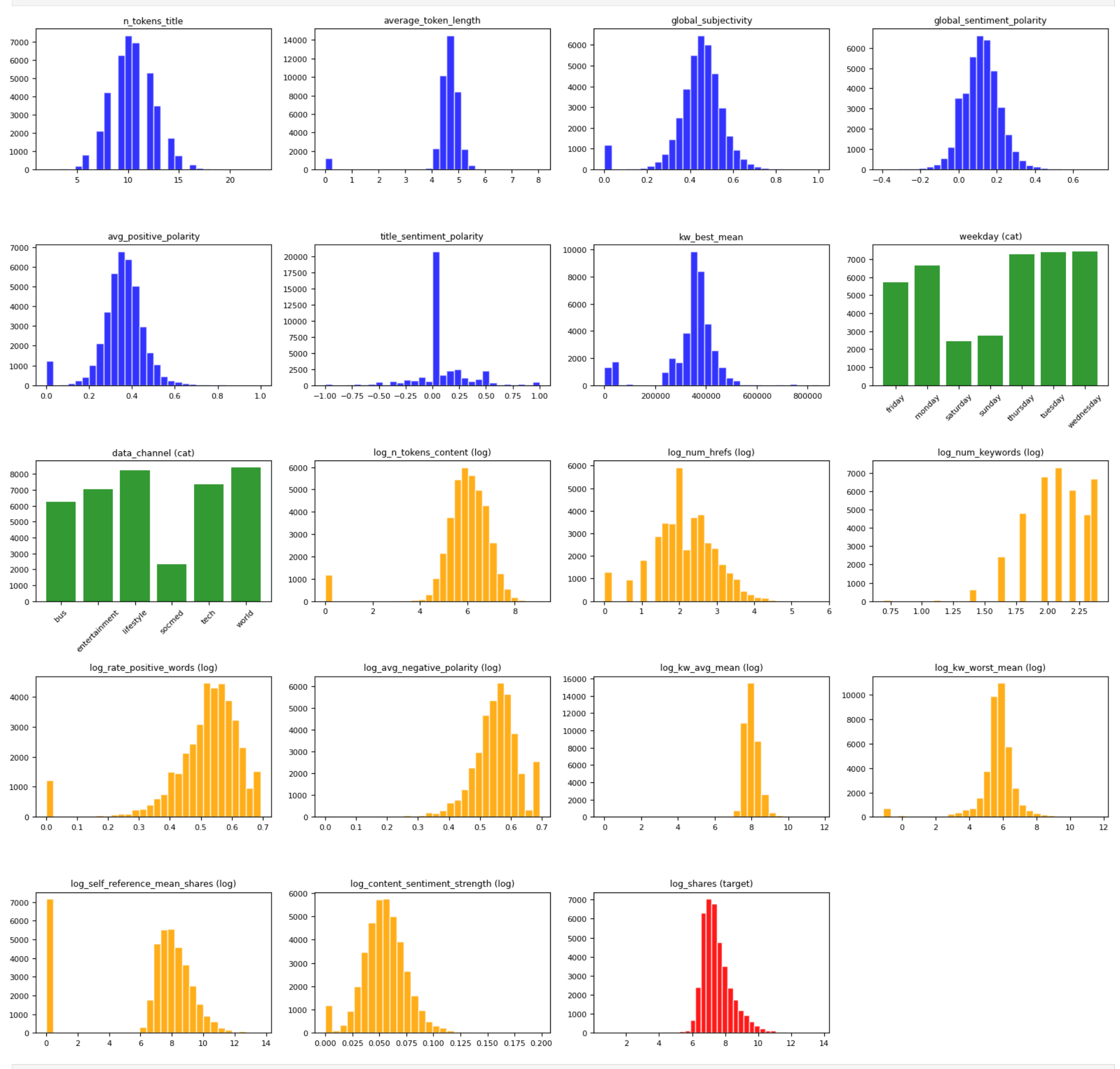
The distributions across weekdays are very similar with comparable IQRs and lots of high outliers on every day. No weekday stands out as clearly better or worse, any differences are small compared to the overall variability.

The exploratory analysis showed that the dataset contains no missing values and includes a large number of heterogeneous predictors. The target variable, number of shares, exhibits a highly right-skewed distribution, which motivated a logarithmic transformation. Several count-based predictors were also strongly right-skewed and were selected for logarithmic transformation. Correlation analysis revealed substantial redundancy among groups of features, leading to feature aggregation and more compact representations. Overall, individual predictors show weak to moderate relationships with the target, suggesting that article popularity depends on complex, non-linear interactions.

Now, we can summarize all the variables on the colorfoul histograms:

In [94]:

```python
# Plot all features in a 4-column grid with color coding:
# - categorical: green
# - target 'log_shares': red
# - logarithmized features (start with 'log_'): orange
# - other numeric features: blue

cols = df.columns.tolist()
n_cols = 4
n_plots = len(cols)
n_rows = math.ceil(n_plots / n_cols)

fig, axes = plt.subplots(n_rows, n_cols, figsize=(4 * n_cols, 3 * n_rows))
axes = axes.flatten()

cat_color = "green"
target_color = "red"
log_color = "orange"
num_color = "blue"

for i, col in enumerate(cols):
    ax = axes[i]
    series = df[col].dropna()

    if col == "log_shares":
        # target (red)
        ax.hist(series, bins=40, color=target_color, edgecolor="white", alpha=0.9)
        ax.set_title(col + " (target)", fontsize=9)
    elif isinstance(df[col].dtype, np.dtype) and df[col].dtype == object:
        # categorical (green) -> bar plot of counts
        vc = series.value_counts().sort_index()
        ax.bar(vc.index.astype(str), vc.values, color=cat_color, alpha=0.8)
        ax.set_title(col + " (cat)", fontsize=9)
        ax.tick_params(axis="x", rotation=45)
    elif col.startswith("log_"):
        # log-transformed features (orange)
        ax.hist(series, bins=30, color=log_color, edgecolor="white", alpha=0.9)
        ax.set_title(col + " (log)", fontsize=9)
    else:
        # other numeric features (blue)
        ax.hist(series, bins=30, color=num_color, edgecolor="white", alpha=0.8)
        ax.set_title(col, fontsize=9)

    ax.tick_params(axis="both", labelsize=8)

# Hide any unused subplots
for j in range(n_plots, len(axes)):
    axes[j].axis("off")

plt.tight_layout()
plt.show()
```



In [96]:

```python
# Print current features in the dataframe
all_columns = df.columns.tolist()

print("All columns (in df):")
for col in all_columns:
    print(f"- {col}")

df.shape
```

```
All columns (in df):
- n_tokens_title
- average_token_length
- global_subjectivity
- global_sentiment_polarity
- avg_positive_polarity
- title_sentiment_polarity
- kw_best_mean
- weekday
- data_channel
- log_n_tokens_content
- log_num_hrefs
- log_num_keywords
- log_rate_positive_words
- log_avg_negative_polarity
- log_kw_avg_mean
- log_kw_worst_mean
- log_self_reference_mean_shares
- log_content_sentiment_strength
- log_shares
```

Out[96]:

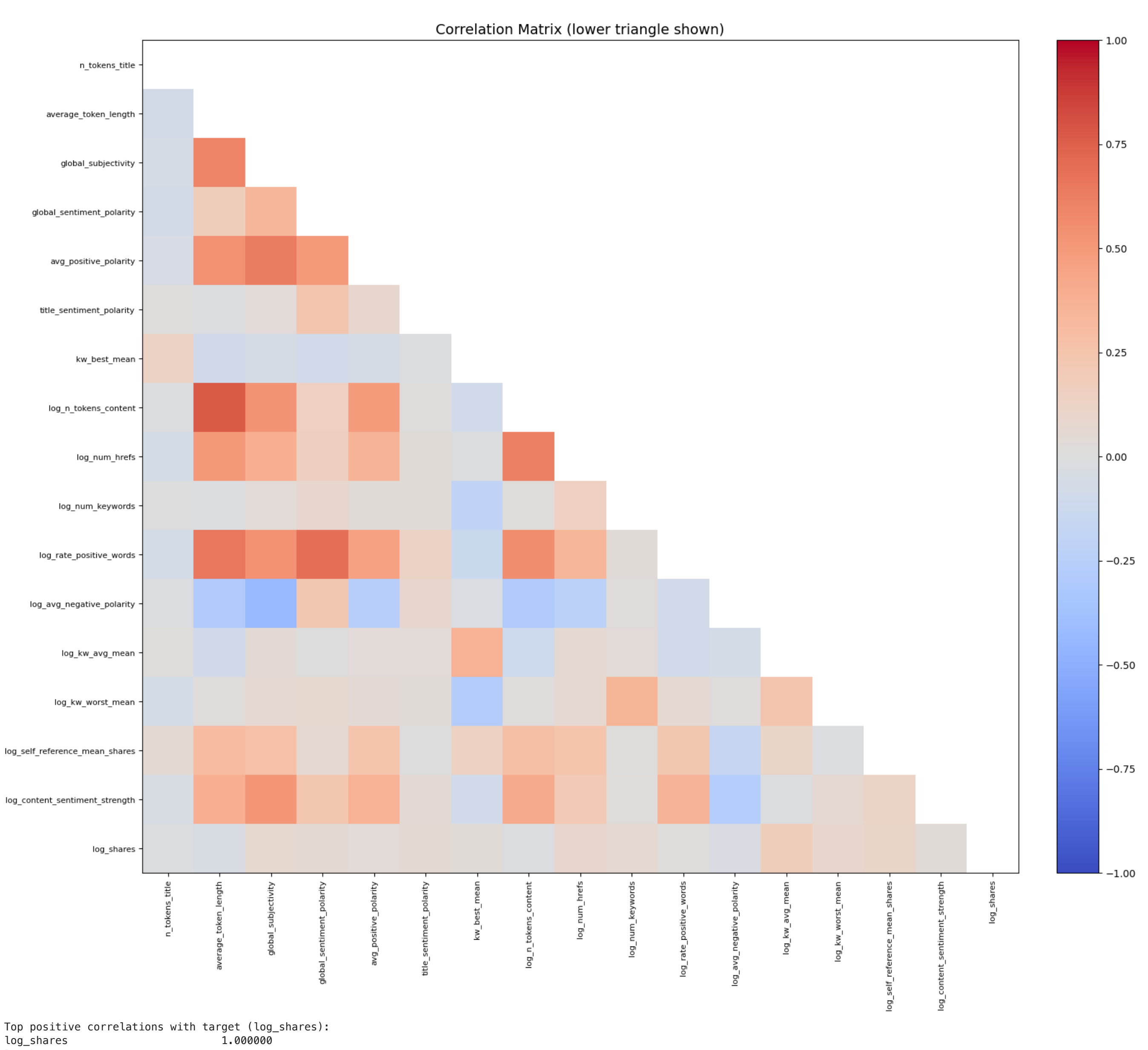
```
(39644, 19)
```

Finally, we create a new correlation heatmap to check the correlations after the transformations. This time we will use different colour pallete.

In [97]:

```python
# Create and display correlation matrix for numeric features, plus a heatmap
corr_matrix = df.corr(numeric_only=True)

# Display the numeric correlation table
display(corr_matrix.round(3))

# Heatmap
mask = np.triu(np.ones_like(corr_matrix.values, dtype=bool))
masked_corr = np.ma.array(corr_matrix.values, mask=mask)

plt.figure(figsize=(16, 14))
cmap = plt.get_cmap("coolwarm")
im = plt.imshow(masked_corr, cmap=cmap, vmin=-1, vmax=1, aspect="auto")
plt.colorbar(im, fraction=0.046, pad=0.04)
plt.xticks(range(len(corr_matrix.columns)), corr_matrix.columns, rotation=90, fontsize=8)
plt.yticks(range(len(corr_matrix.index)), corr_matrix.index, fontsize=8)
plt.title("Correlation Matrix (lower triangle shown)", fontsize=14)
plt.tight_layout()
plt.show()

# Show correlations with the target (use log_shares if present)
target_col = "log_shares" if "log_shares" in corr_matrix.columns else "shares"
print(f"Top positive correlations with target ({target_col}):")
display(corr_matrix[target_col].sort_values(ascending=False).head(10))

print(f"\nTop negative correlations with target ({target_col}):")
display(corr_matrix[target_col].sort_values().head(10))
```

	n_tokens_title	average_token_length	global_subjectivity	global_sentiment_polarity	avg_positive_polarity	title_sentiment_polarity	kw_best_mean	log_n_tokens_content
n_tokens_title	1.000	-0.071	-0.057	-0.072	-0.050	0.000	0.129	-0.017
average_token_length	-0.071	1.000	0.598	0.179	0.540	-0.017	-0.097	0.769
global_subjectivity	-0.057	0.598	1.000	0.339	0.632	0.034	-0.056	0.531
global_sentiment_polarity	-0.072	0.179	0.339	1.000	0.497	0.238	-0.093	0.145
avg_positive_polarity	-0.050	0.540	0.632	0.497	1.000	0.080	-0.059	0.485
title_sentiment_polarity	0.000	-0.017	0.034	0.238	0.080	1.000	-0.022	0.003
kw_best_mean	0.129	-0.097	-0.056	-0.093	-0.059	-0.022	1.000	-0.079
log_n_tokens_content	-0.017	0.769	0.531	0.145	0.485	0.003	-0.079	1.000
log_num_hrefs	-0.064	0.503	0.387	0.152	0.355	0.031	-0.014	0.614
log_num_keywords	-0.003	-0.022	0.038	0.078	0.030	0.029	-0.201	0.018
log_rate_positive_words	-0.065	0.652	0.536	0.690	0.458	0.131	-0.129	0.560
log_avg_negative_polarity	-0.017	-0.297	-0.426	0.226	-0.259	0.085	-0.024	-0.302
log_kw_avg_mean	0.007	-0.097	0.045	-0.003	0.033	0.034	0.371	-0.112
log_kw_worst_mean	-0.068	0.016	0.058	0.075	0.047	0.030	-0.280	0.012
log_self_reference_mean_shares	0.047	0.311	0.277	0.063	0.243	-0.004	0.139	0.289
log_content_sentiment_strength	-0.045	0.388	0.513	0.228	0.372	0.041	-0.093	0.413
log_shares	-0.019	-0.045	0.075	0.049	0.034	0.051	0.028	-0.023



```
Top positive correlations with target (log_shares):
log_shares                        1.000000
log_kw_avg_mean                   0.179287
log_self_reference_mean_shares    0.105474
log_kw_worst_mean                 0.082800
log_num_hrefs                     0.079329
global_subjectivity               0.075174
log_num_keywords                  0.069696
title_sentiment_polarity          0.051038
global_sentiment_polarity         0.048927
avg_positive_polarity             0.034460
Name: log_shares, dtype: float64
Top negative correlations with target (log_shares):
average_token_length             -0.045321
log_avg_negative_polarity        -0.035714
log_n_tokens_content             -0.022876
n_tokens_title                   -0.019048
log_rate_positive_words           0.004188
kw_best_mean                      0.027715
log_content_sentiment_strength    0.029796
avg_positive_polarity             0.034460
global_sentiment_polarity         0.048927
title_sentiment_polarity          0.051038
Name: log_shares, dtype: float64
```

This heatmap is more readable, because of lower number of variables, and because of sharp colours.

Two main clusters of correlated predictors are visible. The first cluster involves sentiment features such as global_subjectivity, global_sentiment_polarity, avg_positive_polarity and title_sentiment_polarity, which are positively correlated with one another. This reflects the fact that sentiment measures derived from the same textual content tend to move in the same direction. The second cluster involves keywords and article statistics, such as log_n_tokens_content, log_num_keywords, and kw_best_mean, which show moderate positive correlations, reflecting relationships between content length, keyword richness, and keyword performance.

Correlations with the target remain relatively weak for all features, although positive sentiment-related variables display slightly stronger correlations than the others. This supports the idea that more positive sentiment in content is modestly associated with higher engagement, but that no single feature provides a strong linear signal for predicting popularity.

Train-Test Split

After the logarithmic transformations are done, it is time to split the dataset. We do a regular 80-20 split.

In [98]:

```python
#Train / test split using log_shares as target
X = df.drop(columns=["log_shares"])
y = df["log_shares"]

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42
)

print("X_train:", X_train.shape)
print("X_test: ", X_test.shape)
print("y_train:", y_train.shape)
print("y_test: ", y_test.shape)
```

```
X_train: (31715, 18)
X_test:  (7929, 18)
y_train: (31715,)
y_test:  (7929,)
```

Encoding of categorical data

We have now 2 categorical variables and we decide that OneHoteEncoder will be the best one. We also considered the TargetEncoder, but for sure we can't use the encoder for ordinal variables. We use the pipeline for the purpose of encoding, in order to prevent the data leakage.

In [197…

```python
# Encode categorical columns
categorical_features = [c for c in ["weekday", "data_channel"] if c in X_train.columns]
numeric_features = [c for c in X_train.columns if c not in categorical_features]

preprocess = ColumnTransformer(
    transformers=[
        ("num", Pipeline(steps=[
            ("scaler", StandardScaler()),
        ]), numeric_features),
        ("cat", OneHotEncoder(drop="first", handle_unknown="ignore"), categorical_features),
    ],
    remainder="drop"
)
```

4. Modelling

We have our dataset split and prepared for the modelling.

For the regression task, we experimented with a diverse set of machine learning algorithms to model nonlinear relationships and evaluate predictive performance.
Random Forest is an ensemble of decision trees trained on bootstrapped samples, whose predictions are averaged to reduce variance.

Gradient Boosting builds trees sequentially, with each new tree correcting the residual errors of the previous ones, improving accuracy over time.

XGBoost is a highly optimized implementation of gradient boosting that incorporates regularization and efficient tree-building strategies to enhance performance and computational speed.

ElasticNet is a linear model that combines L1 and L2 regularization to handle multicollinearity and perform variable selection.

Finally, Support Vector Regression (SVR) maps data into higher-dimensional spaces using kernel functions and fits a margin-based regression line that is robust to outliers and nonlinear patterns.

Gradient Boosting

In [197…

```python
gbr_model = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", GradientBoostingRegressor(random_state=42))
])

gbr_model.fit(X_train, y_train)
y_pred_gbr = gbr_model.predict(X_test)

rmse_gbr = np.sqrt(mean_squared_error(y_test, y_pred_gbr))
r2_gbr = r2_score(y_test, y_pred_gbr)

print(f"Gradient Boosting RMSE: {rmse_gbr:.3f}")
print(f"Gradient Boosting R²: {r2_gbr:.3f}")
```

```
Gradient Boosting RMSE: 0.856
Gradient Boosting R²: 0.145
```

Random Forest

In [ ]:

```python
rf_model = Pipeline(steps=[
("preprocess", preprocess),
("model", RandomForestRegressor(
n_estimators=300,
random_state=42,
n_jobs=-1
))
])

# Fit model
rf_model.fit(X_train, y_train)

# Predict on test set
y_pred_rf = rf_model.predict(X_test)

# Evaluation
rmse_rf = np.sqrt(mean_squared_error(y_test, y_pred_rf))
r2_rf = r2_score(y_test, y_pred_rf)

print(f"Random Forest RMSE: {rmse_rf:.3f}")
print(f"Random Forest R²: {r2_rf:.3f}")
```

```
Random Forest RMSE: 0.857
Random Forest R²: 0.143
```

Elastic Net

In [197…

```python
en_model = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", ElasticNet(
        alpha=0.1,
        l1_ratio=0.5,
        random_state=42,
        max_iter=10000
    ))
])

# Fit model
en_model.fit(X_train, y_train)

# Predict on test set
```

```
y_pred_en = en_model.predict(X_test)

# Evaluation
rmse_en = np.sqrt(mean_squared_error(y_test, y_pred_en))
r2_en = r2_score(y_test, y_pred_en)

print(f"Elastic Net RMSE: {rmse_en:.3f}")
print(f"Elastic Net R²: {r2_en:.3f}")
```

```
Elastic Net RMSE: 0.907
Elastic Net R²: 0.040
```

Support Vector Regression

```
# Baseline SVR model
svr_model = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", SVR(
        kernel="rbf",        # radial basis function kernel (default)
        C=1.0,               # regularization strength
        epsilon=0.1
    ))
])

# Fit model
svr_model.fit(X_train, y_train)

# Predict on test set
y_pred_svr = svr_model.predict(X_test)

# Evaluation
rmse_svr = np.sqrt(mean_squared_error(y_test, y_pred_svr))
r2_svr = r2_score(y_test, y_pred_svr)

print(f"SVR Baseline RMSE: {rmse_svr:.3f}")
print(f"SVR Baseline R²: {r2_svr:.3f}")
```

```
SVR Baseline RMSE: 0.877
SVR Baseline R²: 0.103
```

XGBoost

```
# XGBoost model with the SAME preprocessing
xgb_model = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", XGBRegressor(
        n_estimators=500,        # number of trees
        learning_rate=0.05,      # step size (
        max_depth=6,             # tree depth
        subsample=0.8,           # row sampling
        colsample_bytree=0.8,    # column sampling
        objective="reg:squarederror",
        random_state=42,
        n_jobs=-1
    ))
])

# Fit model
xgb_model.fit(X_train, y_train)

# Predict on test set
y_pred_xgb = xgb_model.predict(X_test)

# Evaluation
rmse_xgb = np.sqrt(mean_squared_error(y_test, y_pred_xgb))
r2_xgb = r2_score(y_test, y_pred_xgb)

print(f"XGBoost RMSE: {rmse_xgb:.3f}")
print(f"XGBoost R²: {r2_xgb:.3f}")
```

```
XGBoost RMSE: 0.858
XGBoost R²: 0.140
```

For each model, we used the same modelling pattern.

We built a regression pipeline with our preprocess step (scaling/encoding) and a regressor. Later fit it on the training data and predicted on the test set. Finally, we computed the insightful metrics.

5. Hyperparameters Tuning

In the next step, we focus on tuning the parameters and finidng the best configuration for each model.

```
def rmse(y_true, y_pred):
    return np.sqrt(mean_squared_error(y_true, y_pred))
```

We build a pipeline that preprocesses the data then trains a regressor.
Later we train a hyperparameter search space and run RandomizedSearchCV optimizing RMSE and also tracking R². For thi purpose, we had to deifne the parameters to test.
Then, the model fits the search on the training data and prints the best hyperparameters plus CV RMSE/R².
Finally it uses the best model to predict the test set and prints test RMSE and R².

Random Forest

Desccription of tuning parameters:

- n_estimators: number of trees in the forest, more trees usually improve stability but increase time,
- max_depth: maximum depth of each tree, deeper trees fit more complex patterns but risk overfitting,
- min_samples_split: minimum samples required to split,
- min_samples_leaf: minimum samples required at a leaf, larger values smooth predictions and reduce variance,
- max_features: number of features considered at each split, smaller values add randomness and reduce overfitting,
- max_samples: fraction/number of samples used to train each tree.

```
rf_pipe = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", RandomForestRegressor(random_state=42, n_jobs=-1))
])

param_dist = {
    "model__n_estimators": [200, 400, 600, 800],
    "model__max_depth": [None, 8, 12, 16, 20, 24],
    "model__min_samples_split": [2, 5, 10],
    "model__min_samples_leaf": [1, 2, 4],
    "model__max_features": ["sqrt", "log2", 0.5],
    "model__max_samples": [None, 0.7, 0.85, 0.95],
}

search = RandomizedSearchCV(
    rf_pipe,
    param_distributions=param_dist,
    n_iter=60,
    scoring={"rmse": make_scorer(rmse, greater_is_better=False), "r2": "r2"},
    refit="rmse",
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=1
)

search.fit(X_train, y_train)

print("Best params:", search.best_params_)

best_idx = search.best_index_
print("Best CV RMSE:", -search.best_score_)
print("Best CV R²:", search.cv_results_["mean_test_r2"][best_idx])

best_rf = search.best_estimator_
y_pred = best_rf.predict(X_test)
print("Test RMSE:", rmse(y_test, y_pred))
print("Test R²:", r2_score(y_test, y_pred))
```

```
Fitting 5 folds for each of 60 candidates, totalling 300 fits
/Users/Weronika/Library/CloudStorage/OneDrive-Osobisty/Pulpit/Data-Science/Year 2 semester 3/Machine Learning 2/ML2_final_project/.venv/lib/python3.13/site-packages/joblib/extern
als/loky/process_executor.py:782: UserWarning: A worker stopped while some jobs were given to the executor. This can be caused by a too short worker timeout or by a memory leak.
  warnings.warn(
Best params: {'model__n_estimators': 600, 'model__min_samples_split': 2, 'model__min_samples_leaf': 4, 'model__max_samples': None, 'model__max_features': 'log2', 'model__max_dept
h': None}
Best CV RMSE: 0.8618504617665235
Best CV R²: 0.14215713510900482
Test RMSE: 0.8506153459386909
Test R²: 0.15592441767815213
```

In the next code snippet, we save the results to the variables, so later on we wll be able to compare the results.

```
rf_search = search
y_pred_rf = y_pred

rf_cv_rmse = -rf_search.best_score_
rf_cv_r2 = rf_search.cv_results_["mean_test_r2"][rf_search.best_index_]

rf_test_rmse = rmse(y_test, y_pred_rf)
rf_test_r2 = r2_score(y_test, y_pred_rf)
```

Gradient Boosting

Description of tuning parameters (new ones):

- learning_rate: shrinks each tree's contribution, lower values need more trees but can generalize better,
- subsample: fraction of training data used per tree.

```
gbr_pipe = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", GradientBoostingRegressor(random_state=42))
])

param_dist = {
    "model__n_estimators": [100, 200, 400],
    "model__learning_rate": [0.03, 0.05, 0.1, 0.2],
    "model__max_depth": [2, 3, 4],
    "model__subsample": [0.7, 0.85, 1.0],
    "model__min_samples_split": [2, 5, 10],
    "model__min_samples_leaf": [1, 2, 4],
}

search = RandomizedSearchCV(
    gbr_pipe,
    param_distributions=param_dist,
    n_iter=30,
    scoring={"rmse": make_scorer(rmse, greater_is_better=False), "r2": "r2"},
    refit="rmse",
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=1
)

search.fit(X_train, y_train)

best_idx = search.best_index_
print("Best params:", search.best_params_)
print("Best CV RMSE:", -search.best_score_)
print("Best CV R²:", search.cv_results_["mean_test_r2"][best_idx])

best_gbr = search.best_estimator_
y_pred = best_gbr.predict(X_test)
print("Test RMSE:", rmse(y_test, y_pred))
print("Test R²:", r2_score(y_test, y_pred))
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
Best params: {'model__subsample': 0.7, 'model__n_estimators': 400, 'model__min_samples_split': 2, 'model__min_samples_leaf': 1, 'model__max_depth': 4, 'model__learning_rate': 0.0
3}
Best CV RMSE: 0.8642358028747579
Best CV R²: 0.13740329352687014
Test RMSE: 0.8530029290152682
Test R²: 0.1511793141525516
```

```
gbr_search = search
y_pred_enet = y_pred

enet_cv_rmse = -gbr_search.best_score_
enet_cv_r2 = gbr_search.cv_results_["mean_test_r2"][gbr_search.best_index_]

enet_test_rmse = rmse(y_test, y_pred_enet)
enet_test_r2 = r2_score(y_test, y_pred_enet)
```

Elastic Net

Description of parameters:

- alpha: overall regularization strength. Higher values shrink coefficients more (simpler model), lower values allow more flexible fits,
- l1_ratio: mix between L1 (lasso) and L2 (ridge) penalties.

```
enet_pipe = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", ElasticNet(random_state=42, max_iter=30000, tol=1e-4))
])

param_dist = {
    "model__alpha": np.logspace(-5, 1, 50),
    "model__l1_ratio": np.linspace(0.05, 1.0, 20),  # no 0
}

search = RandomizedSearchCV(
    enet_pipe,
    param_distributions=param_dist,
    n_iter=30,
    scoring={"rmse": make_scorer(rmse, greater_is_better=False), "r2": "r2"},
    refit="rmse",
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=1
)

search.fit(X_train, y_train)

best_idx = search.best_index_
print("Best params:", search.best_params_)
print("Best CV RMSE:", -search.best_score_)
print("Best CV R²:", search.cv_results_["mean_test_r2"][best_idx])

best_enet = search.best_estimator_
y_pred = best_enet.predict(X_test)
print("Test RMSE:", rmse(y_test, y_pred))
print("Test R²:", r2_score(y_test, y_pred))
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
Best params: {'model__l1_ratio': np.float64(0.05), 'model__alpha': np.float64(0.0005179474679231213)}
Best CV RMSE: 0.8839019263688825
Best CV R²: 0.0976418373995889
Test RMSE: 0.8766727020146969
Test R²: 0.10341828573798806
```

```
enet_search = search
y_pred_enet = y_pred

enet_cv_rmse = -enet_search.best_score_
enet_cv_r2 = enet_search.cv_results_["mean_test_r2"][enet_search.best_index_]

enet_test_rmse = rmse(y_test, y_pred_enet)
enet_test_r2 = r2_score(y_test, y_pred_enet)
```

Support Vector Regression

Description of parameters:

- kernel: shape of the SVR model, rbf captures nonlinear patterns, poly fits polynomial relations, sigmoid is less common,
- C: regularization strength; higher C fits training data more closely, lower C gives smoother models,
- epsilon: larger epsilon ignores more small errors,
- gamma: larger values make the model more sensitive to single points,
- degree: polynomial degree.

```
svr_pipe = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", SVR())
])

param_dist = {
    "model__kernel": ["rbf", "poly", "sigmoid"],
    "model__C": np.logspace(-2, 2, 10),
    "model__epsilon": [0.01, 0.05, 0.1, 0.2, 0.5],
    "model__gamma": ["scale", "auto"],
    "model__degree": [2, 3, 4],  # only used for poly
}

search = RandomizedSearchCV(
    svr_pipe,
    param_distributions=param_dist,
    n_iter=30,
    scoring={"rmse": make_scorer(rmse, greater_is_better=False), "r2": "r2"},
    refit="rmse",
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=1
)

search.fit(X_train, y_train)

best_idx = search.best_index_
print("Best params:", search.best_params_)
print("Best CV RMSE:", -search.best_score_)
print("Best CV R²:", search.cv_results_["mean_test_r2"][best_idx])

best_svr = search.best_estimator_
y_pred = best_svr.predict(X_test)
print("Test RMSE:", rmse(y_test, y_pred))
print("Test R²:", r2_score(y_test, y_pred))
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
Best params: {'model__kernel': 'rbf', 'model__gamma': 'auto', 'model__epsilon': 0.5, 'model__degree': 4, 'model__C': np.float64(0.5994842503189409)}
Best CV RMSE: 0.8695194022562888
Best CV R²: 0.12682828178484312
Test RMSE: 0.8625657443228641
Test R²: 0.1320407749825152
```

```
svr_search = search
y_pred_svr = y_pred

svr_cv_rmse = -svr_search.best_score_
svr_cv_r2 = svr_search.cv_results_["mean_test_r2"][svr_search.best_index_]

svr_test_rmse = rmse(y_test, y_pred_svr)
svr_test_r2 = r2_score(y_test, y_pred_svr)
```

XGBoost

Description of parameters:

- n_estimators: number of trees; more trees can improve performance but take longer,
- learning_rate: shrinkage of each tree's contribution,
- max_depth: maximum depth of each tree,
- subsample: fraction of rows used per tree,
- colsample_bytree: fraction of features used per tree,
- reg_alpha: L1 regularization strength,
- reg_lambda: L2 regularization strength.

```
xgb_pipe = Pipeline(steps=[
    ("preprocess", preprocess),
    ("model", XGBRegressor(
        random_state=42,
        n_jobs=-1,
        tree_method="hist"
    ))
])

param_dist = {
    "model__n_estimators": [200, 400, 600, 800],
    "model__learning_rate": [0.01, 0.05, 0.1, 0.2],
    "model__max_depth": [3, 4, 5, 6, 8],
    "model__subsample": [0.7, 0.8, 0.9, 1.0],
    "model__colsample_bytree": [0.7, 0.8, 0.9, 1.0],
    "model__reg_alpha": [0, 0.01, 0.1],
    "model__reg_lambda": [1.0, 1.5, 2.0],
}

search = RandomizedSearchCV(
    xgb_pipe,
    param_distributions=param_dist,
    n_iter=30,
    scoring={"rmse": make_scorer(rmse, greater_is_better=False), "r2": "r2"},
    refit="rmse",
    cv=5,
    random_state=42,
    n_jobs=-1,
    verbose=1
)

search.fit(X_train, y_train)

best_idx = search.best_index_
print("Best params:", search.best_params_)
print("Best CV RMSE:", -search.best_score_)
print("Best CV R²:", search.cv_results_["mean_test_r2"][best_idx])

best_xgb = search.best_estimator_
y_pred = best_xgb.predict(X_test)
print("Test RMSE:", rmse(y_test, y_pred))
print("Test R²:", r2_score(y_test, y_pred))
```

```
Fitting 5 folds for each of 30 candidates, totalling 150 fits
/Users/Weronika/Library/CloudStorage/OneDrive-Osobisty/Pulpit/Data-Science/Year 2 semester 3/Machine Learning 2/ML2_final_project/.venv/lib/python3.13/site-packages/joblib/extern
als/loky/process_executor.py:782: UserWarning: A worker stopped while some jobs were given to the executor. This can be caused by a too short worker timeout or by a memory leak.
  warnings.warn(
Best params: {'model__subsample': 0.7, 'model__reg_lambda': 1.0, 'model__reg_alpha': 0.01, 'model__n_estimators': 600, 'model__max_depth': 8, 'model__learning_rate': 0.01, 'model
__colsample_bytree': 0.9}
Best CV RMSE: 0.8599416572766044
Best CV R²: 0.1459494829688142
Test RMSE: 0.8501644937630375
Test R²: 0.1568189523513599
```

```
xgb_search = search
y_pred_xgb = y_pred

xgb_cv_rmse = -xgb_search.best_score_
xgb_cv_r2 = xgb_search.cv_results_["mean_test_r2"][xgb_search.best_index_]

xgb_test_rmse = rmse(y_test, y_pred_xgb)
xgb_test_r2 = r2_score(y_test, y_pred_xgb)
```

6. Models comparison

Using the variables we created previously we compare the baseline models first:

```
baseline_table = pd.DataFrame({
    "model": ["Random Forest", "Gradient Boosting", "Elastic Net", "XGBoost", "SVR"],
    "rmse": [rmse_rf, rmse_gbr, rmse_en, rmse_xgb, rmse_svr],
    "r2":   [r2_rf, r2_gbr, r2_en, r2_xgb, r2_svr],
}).set_index("model")

display(baseline_table.round(4))
```

	rmse	r2
model		
Random Forest	0.8573	0.1426
Gradient Boosting	0.8559	0.1453
Elastic Net	0.9070	0.0403
XGBoost	0.8584	0.1405
SVR	0.8768	0.1032

- Overall performance is modest: R² values are low (≈0.04–0.15), so models explain only a small fraction of variance.
- Gradient Boosting is the best of the group (lowest RMSE 0.8559, highest R² 0.1453), but only slightly better than Random Forest and XGBoost.
- Elastic Net performs worst (highest RMSE 0.9070, lowest R² 0.0403).
- We clearly see that the results are extremly poor, we need to check how the situation changes after hyperparameters tuning.

```
def cv_test_row(name, search_obj):
    best_idx = search_obj.best_index_
    cv_rmse = -search_obj.best_score_
    cv_r2 = search_obj.cv_results_.get("mean_test_r2", [np.nan])[best_idx]

    best_model = search_obj.best_estimator_
    y_pred = best_model.predict(X_test)

    return {
        "model": name,
        "cv_rmse": cv_rmse,
        "cv_r2": cv_r2,
        "test_rmse": rmse(y_test, y_pred),
        "test_r2": r2_score(y_test, y_pred),
    }

rows = []
rows.append(cv_test_row("Random Forest", rf_search))
rows.append(cv_test_row("Elastic Net", enet_search))
rows.append(cv_test_row("XGBoost", xgb_search))
rows.append(cv_test_row("SVR", svr_search))
rows.append(cv_test_row("Gradient Boosting", gbr_search))

cv_vs_test = pd.DataFrame(rows).set_index("model")
display(cv_vs_test.round(4))
```

	cv_rmse	cv_r2	test_rmse	test_r2
model				
Random Forest	0.8619	0.1422	0.8506	0.1559
Elastic Net	0.8839	0.0976	0.8767	0.1034
XGBoost	0.8599	0.1459	0.8502	0.1568
SVR	0.8695	0.1268	0.8626	0.1320
Gradient Boosting	0.8642	0.1374	0.8530	0.1512

The table summarizes the performance of five regression models using both crossvalidation metrics and a test set. Two metrics are reported: Root Mean Squared Error (RMSE), where lower values indicate better performance, and R², where higher values indicate better explained variance.

Across all models, results are broadly consistent between cross-validation and test evaluation, indicating that the models generalize reasonably well without severe overfitting or underfitting. Among the evaluated models, XGBoost achieves the best overall performance, with the lowest test RMSE (0.8502) and the highest test R² (0.1568), closely followed by Random Forest and Gradient Boosting, which perform within a similar range. SVR also performs competitively, though with slightly higher error and lower variance explanation. Elastic Net yields the weakest performance, reflecting the limited predictive power of linear models for this dataset, which likely contains nonlinear relationships and interactions.

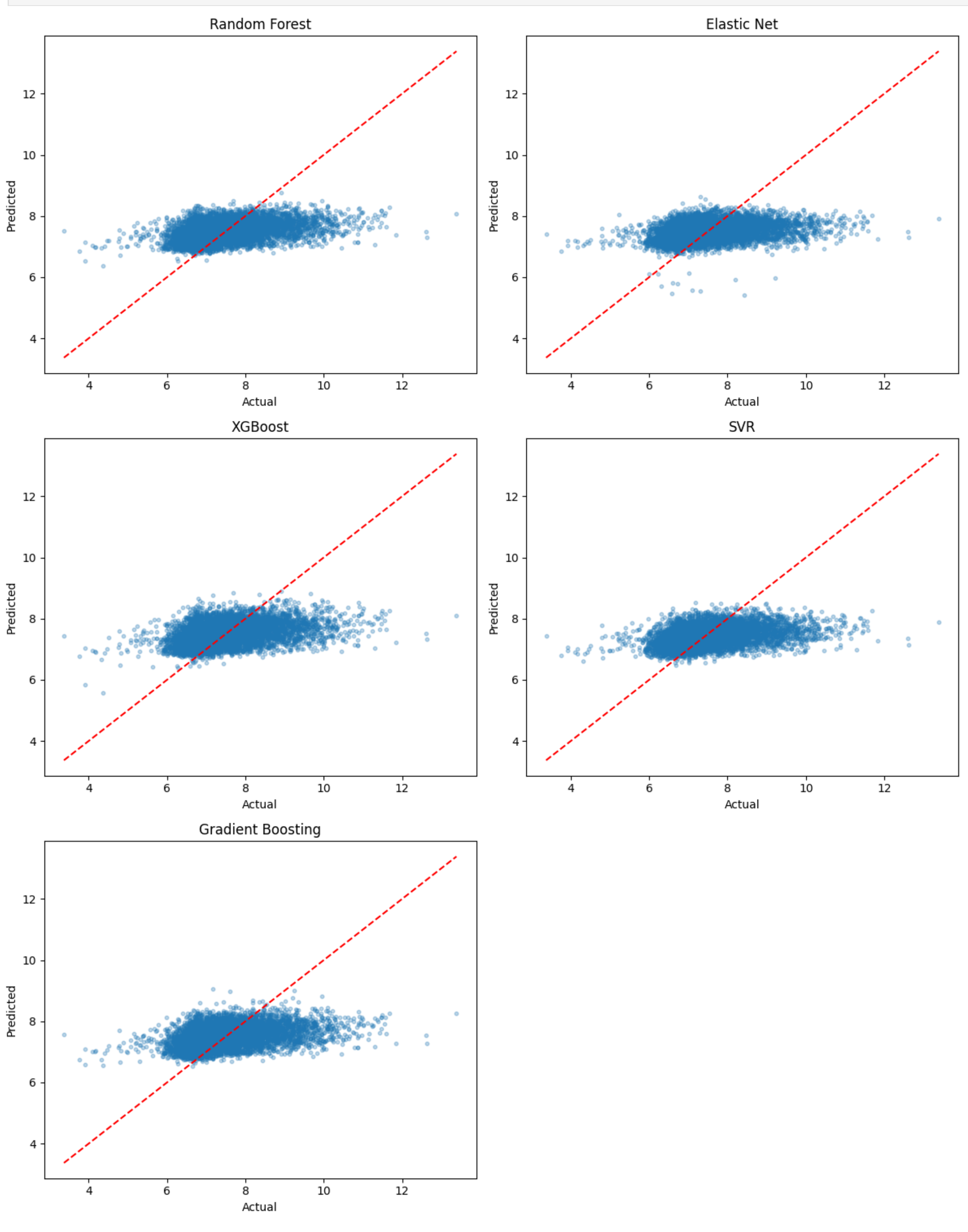
Summing up:

- CV and test metrics are close for all models, so there's no strong overfitting, generalization looks stable.
- Best performers are the tree-based models: XGBoost and Random Forest have the lowest test RMSE (~0.850) and highest test R² (~0.156).
- Gradient Boosting is close behind, SVR is middling, and Elastic Net is clearly weakest (lowest R², highest RMSE).
- Absolute R² values are still low (~0.10–0.16), so even the best models explain only a modest portion of variance.

Hyperparameters tuning didn't really help much in that case. Also investigating different models shows us, that it's not about the modelling, it's rather how insightful are the features selected.

We can also create graphs to see the predicted vs actual results.

```
tuned_models = [
    ("Random Forest", best_rf),
    ("Elastic Net", best_enet),
    ("XGBoost", best_xgb),
    ("SVR", best_svr),
    ("Gradient Boosting", best_gbr),
]

n_cols = 2
n_rows = int(np.ceil(len(tuned_models) / n_cols))
fig, axes = plt.subplots(n_rows, n_cols, figsize=(6*n_cols, 5*n_rows))
axes = np.array(axes).flatten()

y_min, y_max = y_test.min(), y_test.max()

for i, (name, model) in enumerate(tuned_models):
    y_pred = model.predict(X_test)
    ax = axes[i]
    ax.scatter(y_test, y_pred, alpha=0.3, s=10)
    ax.plot([y_min, y_max], [y_min, y_max], color="red", linestyle="--")
    ax.set_title(name)
    ax.set_xlabel("Actual")
    ax.set_ylabel("Predicted")

for j in range(len(tuned_models), len(axes)):
    axes[j].axis("off")

plt.tight_layout()
plt.show()
```



Random Forest

Elastic Net

XGBoost

SVR

Gradient Boosting

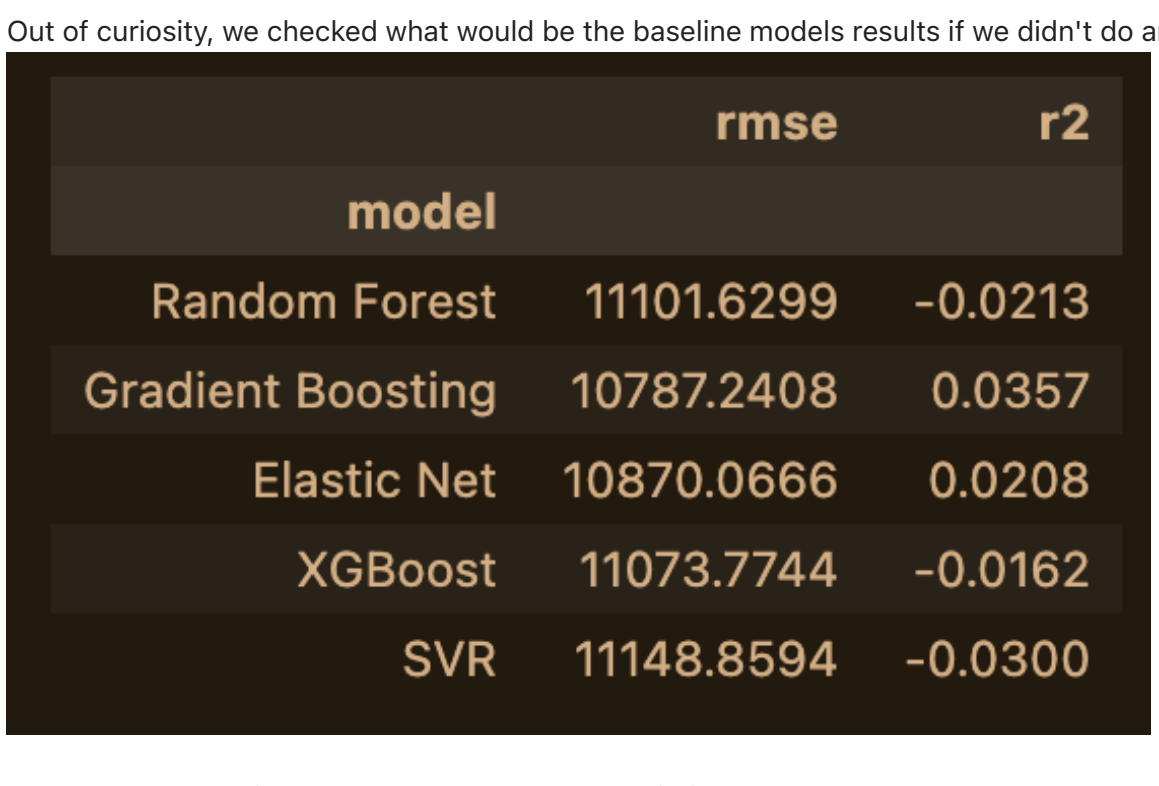
The graphs confirm, the models did not perform very well.

7. Conclusions and what next?

The results turned out to be very poor and not helpful at all. After seeing them, we decided on taking up some actions that we can do to improve the scores. We went through some articles and guides and took some inspiration. What we did:

- decided not to change the weekdays and data_channel variables into the categorical ones and then not encoding them - it didn't bring any meaningful change,
- decided on other choice of features - also didn't bring any meaningful change either,
- tried different types of encoding,
- for the outliers, we already did the logarithmization, however it didn't help with all variables, some of them remained higly-skewed. We decided to apply additional technique of outliers removal - windsorization, but it didn't change the results at all, even after applying extremly narrow ranges,
- we also played with different parameters fo hyperparameters tuning, and with different numbers of iterations. Despite the longer computational time, the changes were neglegable (+/- 0.001 in both, RMSE and R2)

Out of curiosity, we checked what would be the baseline models results if we didn't do any of the feature engineering and what impact our actions actuallly had:



	rmse	r2
model		
Random Forest	11101.6299	-0.0213
Gradient Boosting	10787.2408	0.0357
Elastic Net	10870.0666	0.0208
XGBoost	11073.7744	-0.0162
SVR	11148.8594	-0.0300

The results are saying that these are not real predictions, rather random guesses. But somehow it is a proof, that the actions we undertook did have impact on the dataset and played a role in the predictions, even though the results turned out not to be satisfactory.

Why this happend? The answer is probably because of the very weak correlation of features with target variable form the same beginning. It would be probably more benefitial, if the dataset contained some variables which would be more influental on shares themselves, for example related to some marketing strategies, external events (elections, scandals) or topic relevance to the news.

Another reason might be highly-sked data in most features. Even applying skewness elimination techinques did not have a significant impact and some of the data remained impacted by outliers.

We could also debate, whether the bigger sample would perform better. There are also some suggestions for poor performance, on some websites and forums that the data is inherently noisy, without specific patterns. However, we think that the biggest issue was the weak corrrelation problem.

What next? We think that this project has potential, and an idea for it is changing into the classification model - changing the numbers of shares into some ranges and inspecting the models then. We already have another classification project done and don't want to to quit with poor results, so we decided on an interesting move, which is repeating the actions, but changing the target variable. This will be presented in the second notebook.

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